6. Radioactive decay is the *spontaneous* and random emission of radiation from a radioactive source. Radioactive nuclide $^{131}_{52}\text{Te}$ decays by β emission to form $^{131}_{53}\text{I}$. $^{131}_{53}\text{I}$ is not stable, and decays by β emission to the stable isotope $^{131}_{54}\text{Xe}$. The *half-life* for this second decay is very much longer than that for the decay of $^{131}_{52}\text{Te}$. A sample of pure $^{131}_{52}\text{Te}$ is prepared at time $t = 0$. Fig 6.1 shows the activity *A* of the sample over a period of 10 hours.

***t*/hr**	***A*/Bq**
0	1.00×10^{12}
1	1.94×10^{11}
2	3.95×10^{10}
3	1.08×10^{10}
4	2.66×10^{9}
5	2.27×10^{9}
6	2.12×10^{9}
7	2.12×10^{9}
8	2.12×10^{9}
9	2.12×10^{9}
10	2.12×10^{9}

Fig. 6.1

(a) Explain the meaning of the terms *spontaneous* and *half-life*. [2]

(b) The relation between activity A and time t follows the expression

$$A = A_0 e^{-\lambda t}$$

where A_0 is the initial activity and λ is a constant.

Data from Fig. 6.1 is used to obtain values for ln A. The variation with time t of ln A is plotted on the graph of Fig 6.2.

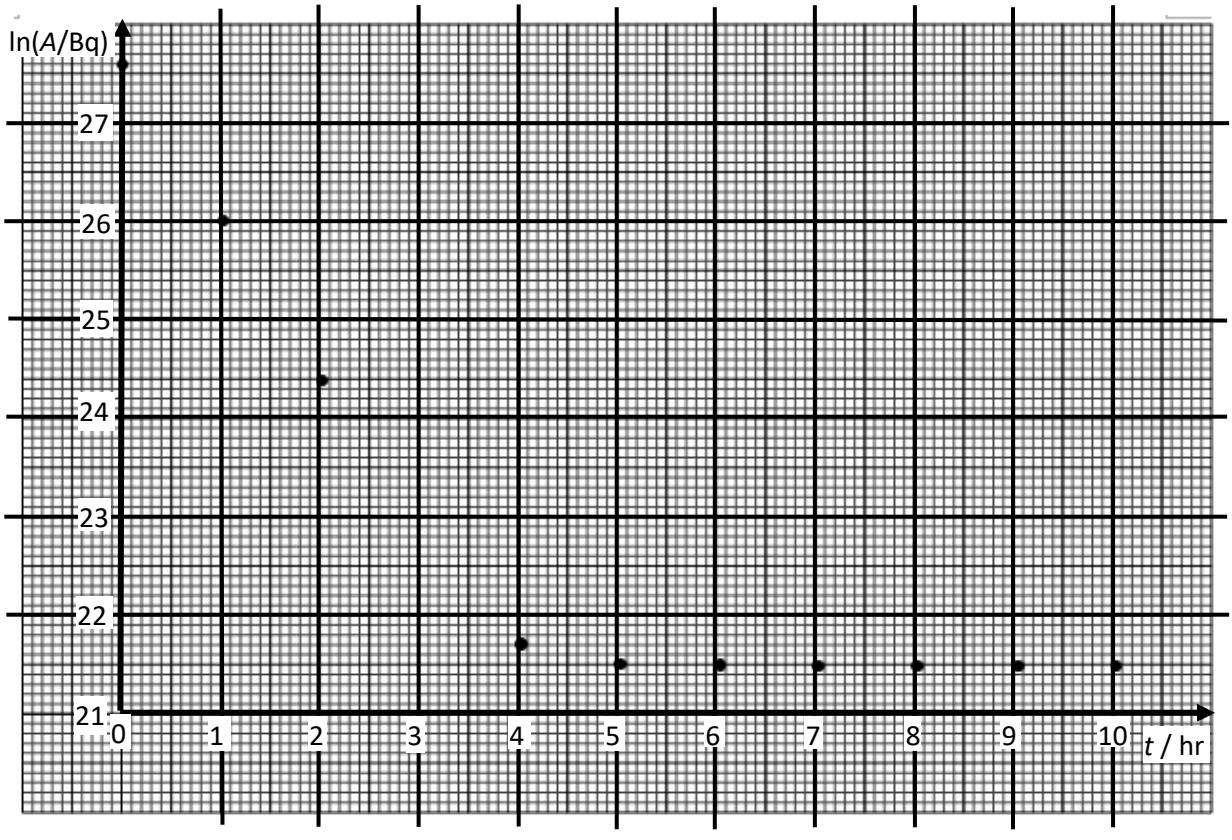

Fig. 6.2

(i) Determine ln A for t = 3 hr. [1]

(ii) On Fig. 6.2,

1. plot the point corresponding to t = 3 hr,
2. draw the line of best fit for all the points. [2]

(iii) Explain the shape of the line drawn in **(ii)**. [2]

(iv) Using the line drawn in **(ii)**, show that the decay constant of $^{131}_{52}Te$ is approximately 1.5 hr^{-1}. [2]

(v) Hence, or otherwise, deduce the initial number of $^{131}_{52}Te$ nuclei. [2]

(vi) Explain why the amount of $^{131}_{53}I$ in the sample first increases and then decreases. [2]

(vii) Assume that all the $^{131}_{52}Te$ has been converted to $^{131}_{53}I$ after about 6 hours. Calculate the half-life of $^{131}_{53}I$. [3]